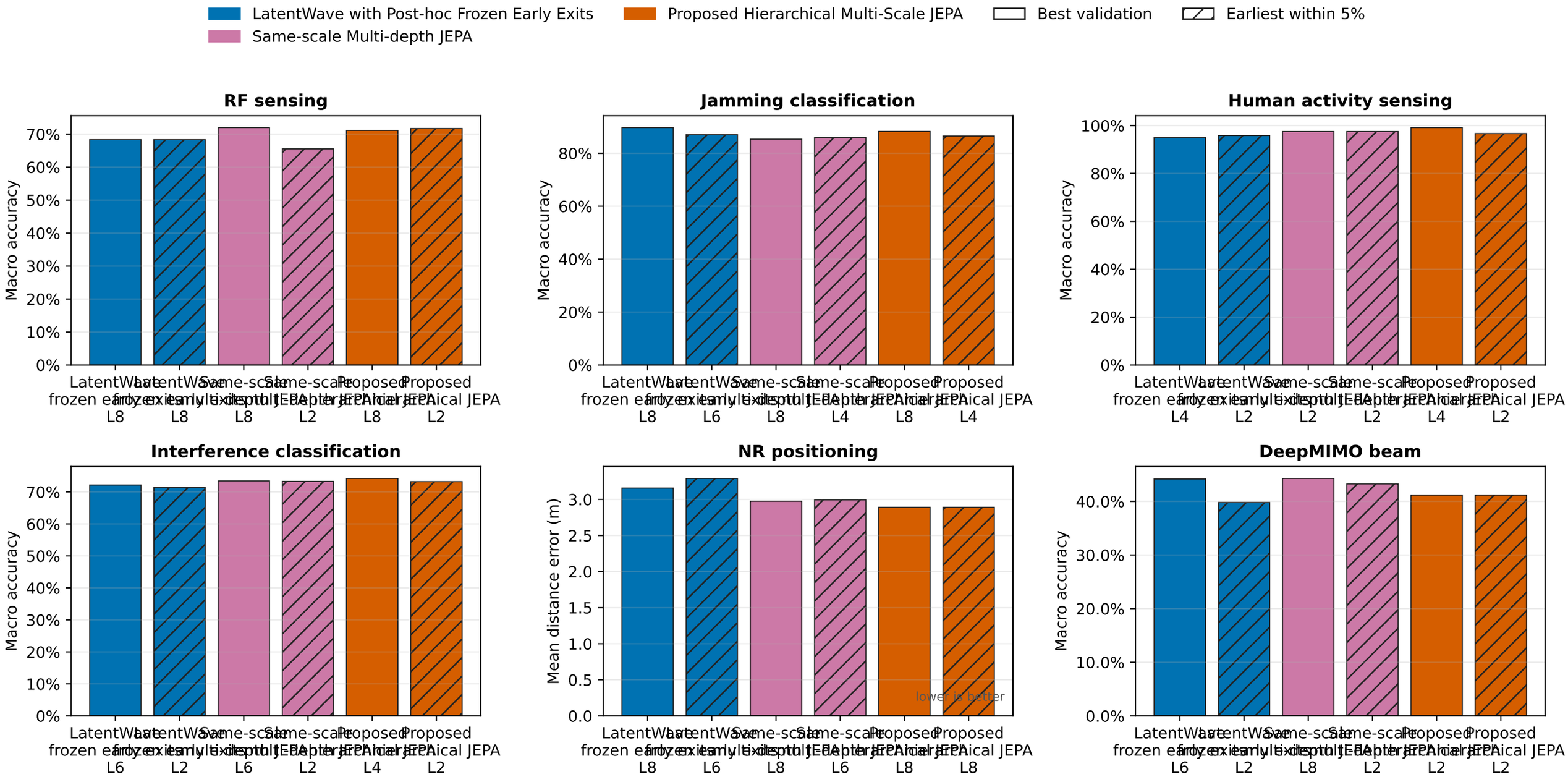
Validation-selected exits and held-out test performance
LatentWave with Post-hoc Frozen Early Exits
Proposed Hierarchical Multi-Scale JEPA
Best validation
Earliest within 5%
Same-scale Multi-depth JEPA
RF sensing
Macro accuracy
0%
10%
20%
30%
40%
50%
60%
70%
LatentWave frozen early exit L8
LatentWave early exit L8
Same-scale JEPA L8
Same-scale JEPA L2
Proposed JEPA L8
Proposed Hierarchical JEPA L2
Jamming classification
Macro accuracy
0%
20%
40%
60%
80%
LatentWave frozen early exit L8
LatentWave early exit L6
Same-scale JEPA L8
Same-scale JEPA L4
Proposed JEPA L8
Proposed Hierarchical JEPA L4
Human activity sensing
Macro accuracy
0%
20%
40%
60%
80%
100%
LatentWave frozen early exit L4
LatentWave early exit L2
Same-scale JEPA L2
Same-scale JEPA L2
Proposed JEPA L4
Proposed Hierarchical JEPA L2
Interference classification
Macro accuracy
0%
10%
20%
30%
40%
50%
60%
70%
LatentWave frozen early exit L6
LatentWave early exit L2
Same-scale JEPA L6
Same-scale JEPA L2
Proposed JEPA L4
Proposed Hierarchical JEPA L2
NR positioning
Mean distance error (m)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
lower is better
LatentWave frozen early exit L8
LatentWave early exit L6
Same-scale JEPA L8
Same-scale JEPA L6
Proposed JEPA L8
Proposed Hierarchical JEPA L8
DeepMIMO beam
Macro accuracy
0.0%
10.0%
20.0%
30.0%
40.0%
LatentWave frozen early exit L6
LatentWave early exit L2
Same-scale JEPA L8
Same-scale JEPA L2
Proposed JEPA L2
Proposed Hierarchical JEPA L2
Exit labels show the selected encoder depth; selection never uses the test split.